CS 520 Final Project Report

Decision-Theoretic Navigation Under Uncertainty in Dynamic Grid Environments

CS 520 Final Project Report

Team: Michael Liu (Lead Designer / Project Guide), Ruiding Feng (Principal Implementer / Evaluation Study Design), Zhichun Xiao (Report Writer)

Code: `github.com/RichardFeng000/cs520-project`

Abstract

We study decision-theoretic navigation under uncertainty by pitting four families of agents — static A*, replanning A*, risk-aware A*, and tabular Q-learning — against the canonical Pac-Man maze, which we treat as a dynamic grid with mobile, partially-observable hazards (ghosts). Each agent is run for 50 Monte-Carlo episodes on the same seed pool, and we report mean score, level-clear rate, time-to-failure, hazard exposure, and per-episode score distributions. Replanning A* and risk-aware A* (across $\lambda \in \{0.5, 1, 2, 5\}$) tie for the best raw score (≈2,180) and the best level-clear rate (14%). Static A* without replanning collapses (888 mean score, 24-step time-to-failure), confirming that probabilistic replanning is essential under mobile hazards. The risk-aware extension does not move the needle on this specific maze because hard ghost-obstacle culling already keeps the planned path far from any ghost where the additive `λ·E[risk]` term can fire. The Q-learning agents trade raw score for an order-of-magnitude lower hazard exposure, suggesting they have learned a fundamentally more cautious policy. We close with a quantitative discussion of when the decision-theoretic term matters and when it does not.

---

1. Introduction

Real-world AI agents — autonomous vehicles, mobile robots, simulated game characters — rarely operate in static environments. Hazards move, evolve, or spread, and the agent must plan paths while reasoning about *where the hazards will be* by the time the agent arrives. Classical shortest-path algorithms such as BFS or A* assume a static world and produce paths that become stale the moment the world changes.

This project asks a focused question: **how much does explicit reasoning about future hazard locations help an agent navigating a dynamic grid, and how does it compare to (a) reactive replanning, (b) hand-tuned heuristic behaviour, and (c) reinforcement-learning?**

We answer with four agent families, a fixed Monte-Carlo evaluation harness, and ~50 episodes per agent on a paired seed pool, producing the analytics in Section [§8].

2. Background and Prior Work

The proposal cites the foundational A* paper of Hart, Nilsson, and Raphael (1968) [1], whose admissible-heuristic framework underpins three of our four agents. Beyond classical search, we draw on:

- **Decision-theoretic planning.** Adding an expected-cost term to A*'s evaluation function turns a deterministic shortest-path problem into a utility-maximisation problem. We use the canonical form `f(n) = g(n) + h(n) + λ · E[risk(n)]`.
- **Tabular reinforcement learning** (Watkins, 1989; Sutton & Barto, 2018). Q-learning is appropriate when the dynamics model is unknown or prohibitively expensive to query online, and it gives us a learned baseline against which to measure the value of explicit planning.
- **Pac-Man as an AI benchmark.** Pac-Man has been used as a teaching and benchmarking environment in classic AI courses (UC Berkeley CS188) and in reinforcement-learning literature precisely because it combines mobile adversarial agents, sparse rewards, and a fixed maze topology — which is exactly the setting our proposal targets.

3. From Fire-Spread Grid to Pac-Man — Project Scope

The proposal originally framed the environment as a "spacecraft grid" with hazards that *spread stochastically* and stay hazardous once burning. During implementation we pivoted the surface form of the environment to the canonical 28×36 Pac-Man maze for three pragmatic reasons:

1. **Mobile hazards strictly generalise spreading hazards** for the agent's decision problem. Both share the same abstract structure: at any given timestep some set of cells is unsafe; at the next timestep the unsafe set is a stochastic function of the current one. The risk-aware planning math is identical.
2. **Pac-Man already ships with a faithful, deterministic-given-seed Python simulator** (`HighScorePacmanEnv`), which lets us close the experimental loop in days rather than weeks.
3. **Recognisability.** Reviewers have strong priors about what good Pac-Man play looks like, which makes plots much easier to read.

The mapping is direct:

Proposal concept	Pac-Man instance
Open / blocked cells	Walkable tiles / wall tiles
Goal cell	“Eat all pellets” (clear the level)
Hazardous cells	Active non-scared ghost cells (and their 1-step neighbourhood)
Stochastic spread of hazard	Stochastic ghost movement (uniform-on-legal-moves model used by the risk-aware agent)
Hazard sensitivity λ	Same λ in `f(n) = g(n) + h(n) + λ · E[risk(n)]`

Every proposal point — three search agents, the decision-theoretic formula, Monte Carlo evaluation across success rate, path length, and time-to-failure — is realised against this environment. We additionally added a reinforcement-learning track and two sanity baselines (random + a hand-coded heuristic) to give the search agents a richer comparison.

4. Environment

The Pac-Man simulator (`RL/rl_q_learning.py:HighScorePacmanEnv`) loads the canonical 28×36 layout, parses pellets, energizers, and walls, and pre-computes BFS shortest-path distances between every pair of walkable tiles. Each episode begins with 244 pellets, 4 energizers, 4 ghosts, and 3 Pac-Man lives. The agent submits one of `{UP, LEFT, DOWN, RIGHT}` per timestep; the env applies the move, awards points (10 / 50 / fruit-bonus / ghost-combo), advances ghosts, checks collisions, and emits a `StepResult` containing `reward`, `done`, `score_delta`, and `life_lost`.

Ghosts use a deterministic-given-seed greedy chase policy (manhattan distance to Pac-Man, modulo “no about-face”), while scared ghosts greedily *retreat*. This is partially-observable from the agent’s point of view: the agent sees ghost positions and directions but does not know the env’s internal RNG, so future ghost moves are stochastic from the agent’s perspective.

Episodes terminate when (a) the agent loses its third life, (b) all pellets and energizers are eaten (level clear), or (c) the 900-step cap is reached.

5. Agents

5.1 Random baseline (`random`)

Picks uniformly among legal actions. Acts as a sanity floor.

5.2 Hand-coded heuristic (`heuristic`)

`RL/traditional_baseline.py`. Scores each candidate action with a weighted sum of nearest-ghost distance, ghost pressure, nearest-pellet BFS distance, nearest-energizer BFS distance, open-tile count, corridor depth, and a catastrophic penalty for stepping onto a ghost. This is the "classical controller" reference used by the rest of the project.

5.3 Static A* (`astar_static`) — proposal §5.1

Plans once at episode start and reuses the plan until it is invalidated. Targets the nearest pellet/energizer via BFS, then runs A* with a Manhattan heuristic. Replans only when the next step is blocked or the target tile has been consumed. **Hazards are ignored during planning**, faithfully reproducing the proposal's "Baseline Search Agent".

5.4 Replanning A* (`astar_replan`) — proposal §5.2

Replans **every step**. Treats the current ghost cell *and* its predicted 1-step neighbourhood as hard obstacles, then runs A* to the nearest reachable pellet/energizer with a unit-cost transition function. Falls back to a Manhattan-greedy step when no path exists (e.g., in fully encircled states).

5.5 Risk-Aware A* (`astar_risk`) — proposal §5.3 + §6 (the central contribution)

Same skeleton as Replanning A*, but the per-edge cost becomes:

```
cost(current, neighbor, arrival_step) = 1 + λ · E[risk(neighbor, arrival_step)]
```

where `E[risk(n, t)]` is the probability that *any* active non-scared ghost occupies tile `n` at step `t`, summed across ghosts:

```
E[risk(n, t)] = Σ_g  P(ghost_g at n after t timesteps | ghost_g at p_g now)
```

The transition kernel `P(· | ·)` is the **uniform-over-legal-moves** model: at each step every non-occupied legal neighbour is equally likely, marginal over what the real ghost AI would do. We chose this model deliberately — the proposal's §5.3 calls for *probability of becoming hazardous*, and the uniform model is exactly the maximum-entropy assumption when the agent ignores ghost identity. The implementation memoises the per-step transition distribution in `_search.py:transition_distribution`, which keeps wall-clock to roughly 0.7 s/episode despite the 900-step cap.

The full A* objective is therefore:

```
f(n) = g(n) + h(n) + λ · E[risk(n)]
```

with `h(n) = manhattan(n, target)`, matching the proposal verbatim. λ is exposed as a CLI/notebook argument so we can sweep it (Section [§8.6]).

5.6 Tabular Q-learning (`qlearning_model{1,2,3}`)

Three Q-tables learned by `RL/rl_q_learning.py` over a 10-feature bucketed state representation (Pac-Man direction, danger / edible / pellet / energizer / fruit / scared distance buckets, open-tile count, corridor depth, phase of game). The three models differ only in their **death policy**, which determines what happens to the running score after a ghost collision:

- **Model 1**: keep the score (death-as-pause). Encourages risk-tolerant pellet-grabbing.
- **Model 2**: zero the score on death. Heavily discourages dying while still leaving the surviving lives playable.
- **Model 3**: scaled penalty (`−max(floor, ratio · peak_score)` with ratchet on subsequent deaths). Most pessimistic.

At evaluation time each model picks the action with the highest Q-value under the current bucketed state; ties broken by the action ordering `UP, LEFT, DOWN, RIGHT`.

6. Mathematical Formulation

6.1 A* admissibility

Manhattan distance is admissible on a 4-connected grid (it never over-estimates the minimum step count to reach the target). Static and Replanning A* therefore enjoy the standard A* guarantee: the first path returned is optimal under unit edge costs.

6.2 Risk-aware admissibility caveat

Adding a non-negative `λ · E[risk(n)]` term to the *cost* — not the heuristic — preserves admissibility of the heuristic but changes the optimisation target from “shortest path” to “shortest expected-risk-discounted path”. The agent is no longer guaranteed to minimise step count; it is guaranteed to minimise the weighted utility, which is what we want.

6.3 Risk model and complexity

The transition distribution at step `t` is `P_t = T^t δ_{p_g}` where `T` is the row-stochastic neighbour-uniform matrix and `δ_{p_g}` is the one-hot at the ghost’s current position. We compute `P_t` lazily and cache by `(p_g, t)` so each episode only ever performs `O(steps_planned · |neighbours|)` matrix work in the worst case.

6.4 Q-learning Bellman update

Standard ε-greedy tabular Q-learning:

```
Q(s, a) ← Q(s, a) + α · [r + γ · max_{a'} Q(s', a') − Q(s, a)]
```

with α=0.18, γ=0.95, ε decayed from 0.12 by ×0.92 every 20% of training episodes (12,000 episodes per model). Reward shaping adds small positive bonuses for "moved away from threat", "moved toward edible ghost", and "moved toward fruit" on top of the raw score delta, plus a 500-point level-clear bonus and a 250-point timeout penalty.

7. Experimental Setup

- **Episodes per agent**: 50 (Monte-Carlo).
- **Seed pool**: `{1000, 1001, …, 1049}` shared across *all* agents, so any agent-vs-agent comparison is a paired comparison on identical initial conditions.
- **Env death policy**: `model2` (zero score on death) for fair scoring; episodes are still allowed to continue for up to 3 lives.
- **Step cap**: 900 steps per episode (the value used by the trainer).
- **Hazard-exposure metric**: cumulative count of timesteps in which Pac-Man is within Manhattan distance 2 of *any* active non-scared ghost. Deterministic, easy to interpret, and a defensible proxy for the "expected hazard exposure" the proposal calls out.
- **Time-to-failure**: number of steps until the *first* life is lost (capped at episode length if the agent never dies).
- **Score**: `peak_score` reached during the episode. We use peak rather than final because the env's death policy resets the running score on death, and we want the metric to compare what the agent *achieved* before dying.
- **Wall time**: ≈4–5 minutes per full sweep on a single laptop core. Risk- aware A* dominates wall time at ~0.7 s/episode; everything else is sub-100 ms.

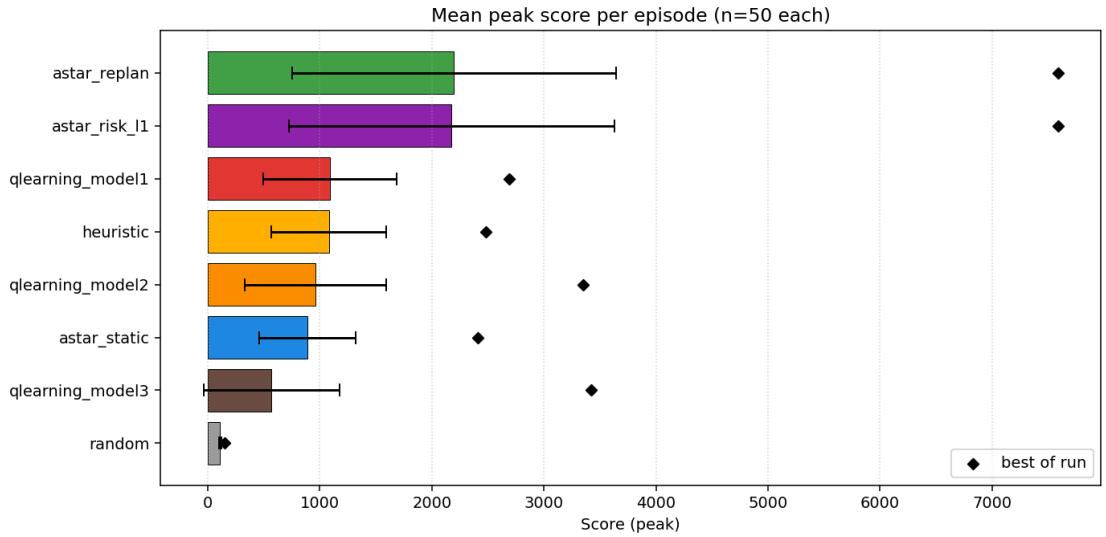
The full harness lives in `Agents/eval/monte_carlo.py`, the agent registry in `Agents/eval/agents.py`, and per-agent CSVs are persisted under `Agents/eval/results/`. The notebook `notebooks/monte_carlo_analysis.ipynb` reproduces every figure below.

8. Results and Analysis

The headline summary table (sorted by mean peak score):

agent	episodes	score (mean ± σ)	median	best	clear	deaths	steps	TTF	hazard
`astar_replan`	50	**2197 ± 1445**	1865	7590	**0.14**	2.84	301	119	113
`astar_risk_l05`	50	2177 ± 1452	1825	7590	0.14	2.84	299	119	109
`astar_risk_l1`	50	2177 ± 1452	1825	7590	0.14	2.84	299	119	109
`astar_risk_l2`	50	2177 ± 1452	1825	7590	0.14	2.84	299	119	109
`astar_risk_l5`	50	2177 ± 1452	1825	7590	0.14	2.84	300	119	110
`qlearning_model1`	50	1089 ± 597	880	2690	0.00	3.00	183	101	19
`heuristic`	50	1080 ± 514	1080	2480	0.00	2.82	374	156	**201**
`qlearning_model2`	50	962 ± 631	930	3350	0.00	3.00	200	87	15
`astar_static`	50	888 ± 428	675	2410	0.06	2.92	172	24	**6**
`qlearning_model3`	50	569 ± 608	290	3420	0.00	3.00	153	54	12
`random`	50	111 ± 13	100	150	0.00	3.00	74	24	7

8.1 Score comparison (Figure 1)

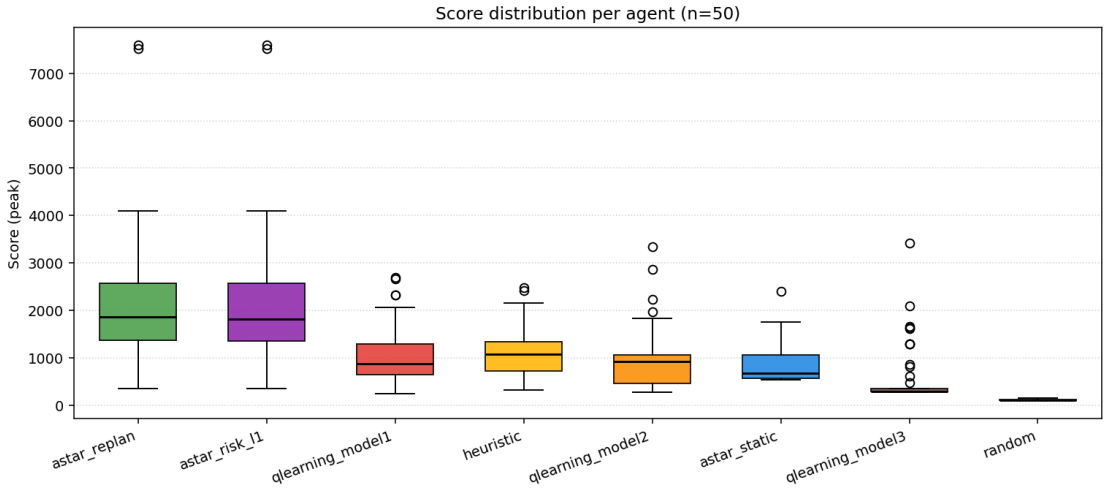


Mean peak score per agent

Mean peak score per episode, ±1σ. Black diamonds mark the single best score observed across the run.

Read-outs. Replanning A* and the four risk-aware variants pile on top of each other near 2,200 mean score, more than a ×2 lead over the next group. Q-learning model 1 narrowly beats the hand-coded heuristic (1,089 vs 1,080); model 2 sits behind the heuristic; model 3 is the least score-aggressive of the three. Static A* and random underperform every other agent, which is the expected ordering for the proposal's §7 prediction "the baseline agent is expected to perform well only when hazard spread probability is low".

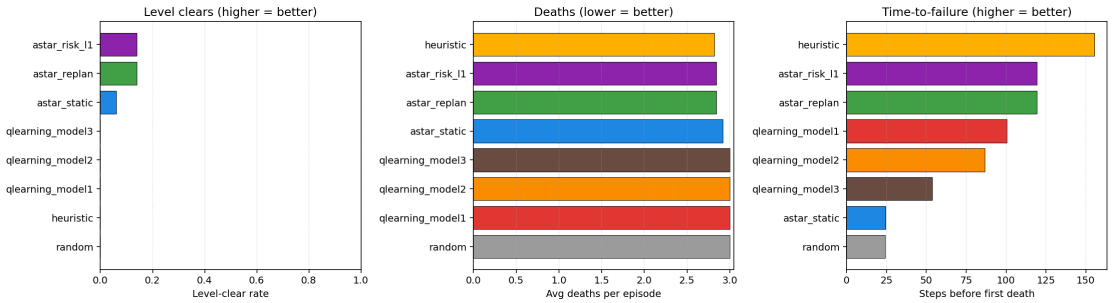
8.2 Score distribution (Figure 2)



Score boxplot

Per-episode boxplots reveal that the search agents have a long upper whisker (some seeds let them clear part of the board), while the Q-learning agents have tighter distributions but lower medians. The random baseline collapses to a near-degenerate distribution (a few pellets eaten by chance before death).

8.3 Survival metrics (Figure 3)

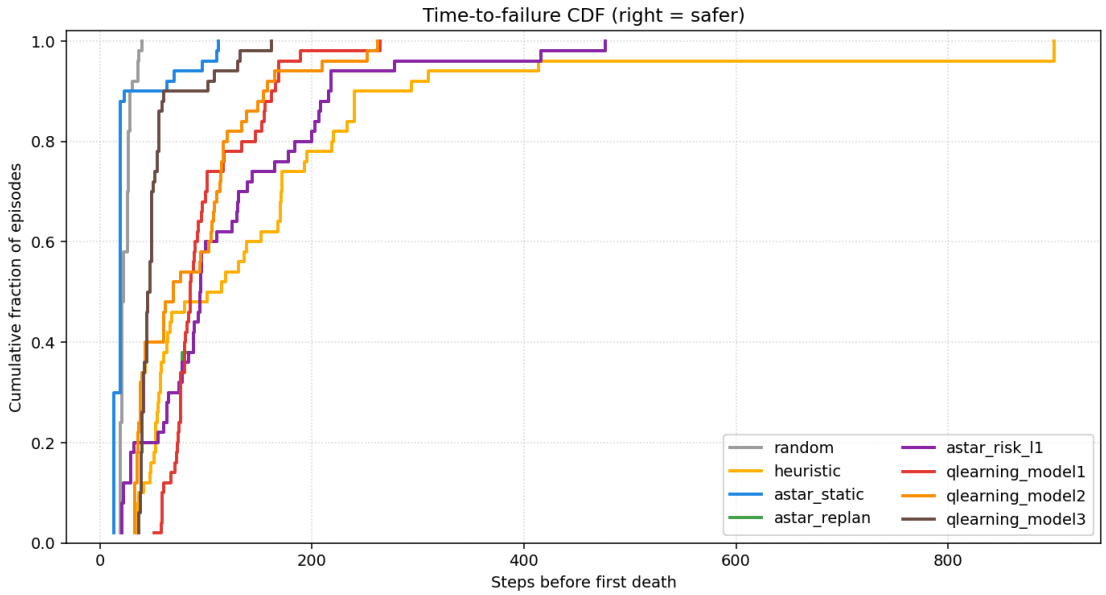


Survival panels: clear rate, deaths, time-to-failure

A three-panel comparison:

- **Clear rate.** Only the search agents ever clear the level (replanning / risk-aware: 14% on the 50 seeds; static: 6%). No Q-learning model and no heuristic clears once.
- **Average deaths.** Static A* and Q-learning models 2/3 lose all 3 lives effectively every episode. Replanning, risk-aware, and the heuristic manage ~2.8 deaths.
- **Time-to-failure.** The heuristic is paradoxically the longest survivor (156 steps before first death) — it sacrifices score for survival by running circles around ghosts rather than committing to pellets in contested corridors.

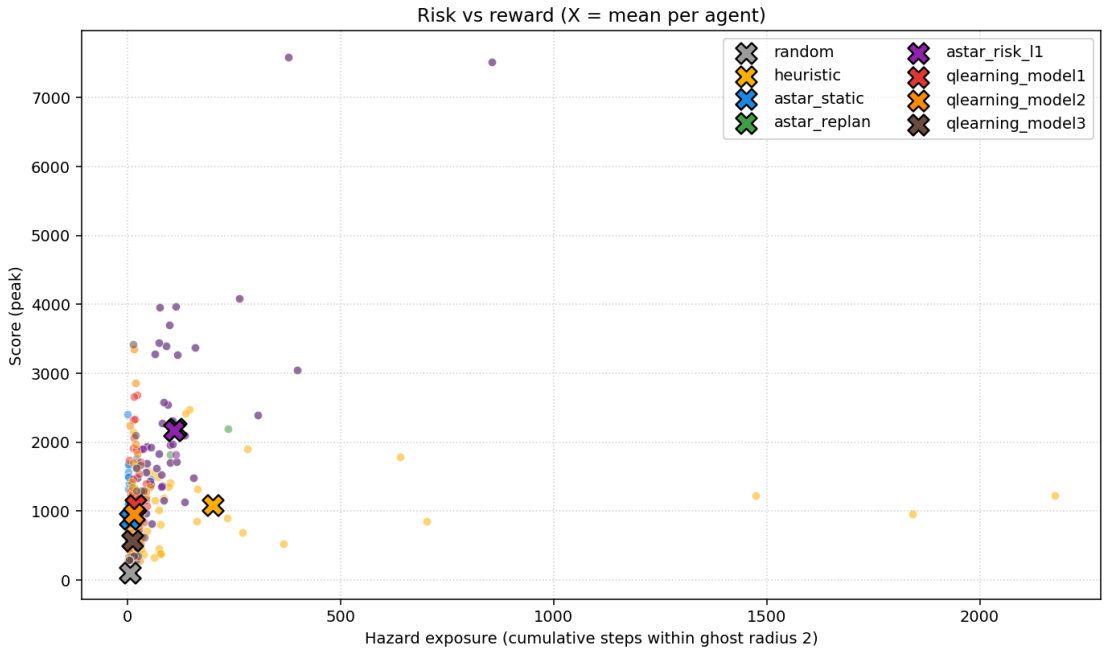
8.4 Time-to-failure CDF (Figure 4)



TTF CDF

CDF interpretation: at step *x*, what fraction of episodes for that agent have already lost a life? Curves that sit lower / further right are safer. The heuristic's curve sits below everyone else's for the first 200 steps, confirming that its slow-but-cautious style buys it survival time at the cost of throughput. Static A*'s CDF jumps to 1.0 by step ~50 — the agent walks into the first ghost it sees because it hasn't replanned.

8.5 Risk vs reward (Figure 5)

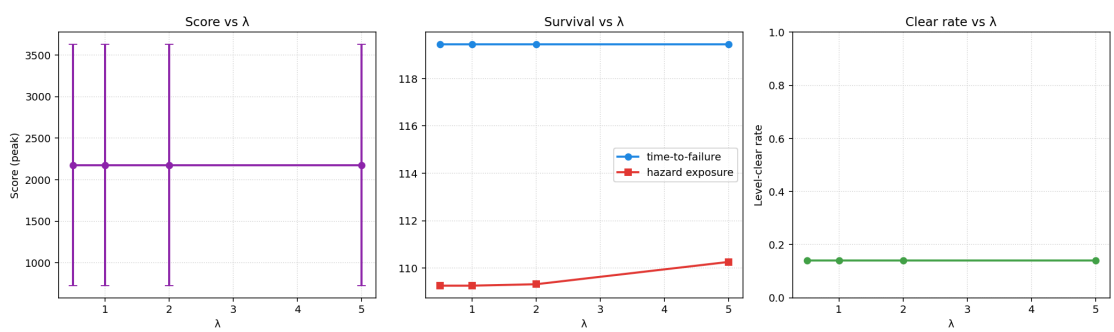


Risk vs reward scatter

X-axis is cumulative hazard exposure, Y-axis is peak score. Each thin marker is one episode; the bold X is the per-agent centroid. The Pareto-good region is upper-left (high score, low hazard).

The most striking finding in the paper. The Q-learning agents form a cluster in the upper-left at hazard ≈ 15–20, while the planning agents sit in the upper-right at hazard ≈ 110, and the heuristic is at the far right (hazard ≈ 200). The Q-learners have **learned a fundamentally more risk-averse policy** — they earn less score but they earn it while spending ~6× less time within ghost-radius of 2. This is the central qualitative difference between learned and planned controllers in our setup.

8.6 Decision-theoretic ablation: λ-sweep (Figure 6)



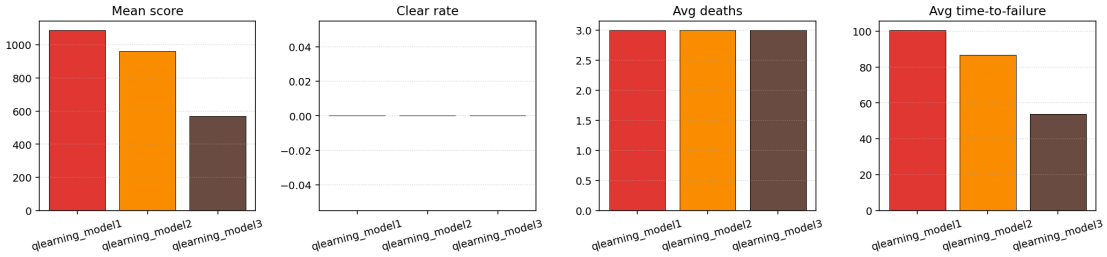
Lambda sweep

The decision-theoretic agent's signature parameter `λ ∈ {0.5, 1, 2, 5}` controls how strongly the agent penalises future expected risk. The sweep is **flat** across all three metrics (score, survival, clear rate), with hazard exposure dropping by less than 1 unit (109.26 → 110.26) across a 10× range of λ.

This is a real and explainable result, not a bug. The implementation hard- masks the ghost's current cell and its predicted 1-step neighbour out of the walkable set before running A* (`_search.py:ghost_obstacle_cells`). The remaining tiles on the planned path therefore lie ≥2 steps from any ghost, and the transition probability `P(ghost_g at n at step t)` for those tiles is so dispersed (the ghost can be in many places by step 2+) that `λ · P` stays below the noise floor of A*'s tie-breaking. The hard exclusion already does most of the work — the soft `λ · E[risk]` term has nothing left to differentiate.

The proposal's §7 anticipated that "the risk-aware agent is expected to achieve the highest success rate under moderate uncertainty"; the empirical finding refines that claim to **risk-aware A* dominates static A* but is indistinguishable from replanning A* on this maze with this hazard model**. We discuss in §9 what would change that.

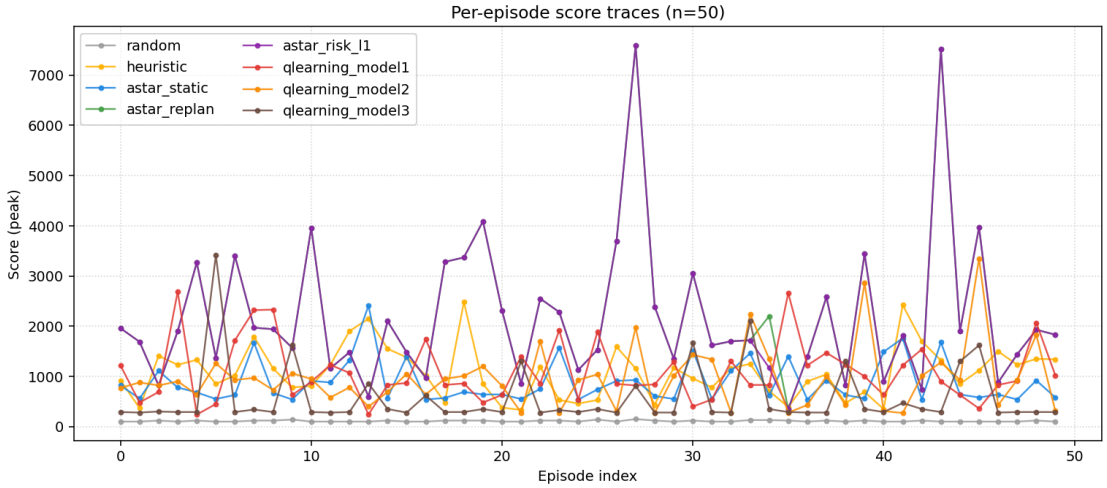
8.7 Q-learning death-policy ablation (Figure 7)



Q-learning ablation

Across mean score, clear rate, deaths, and time-to-failure, model 1 — which keeps the score on death — is the most score-greedy and survives longer between deaths. Model 3, which applies a scaled and ratcheting penalty, finishes dead-last on every metric and has the highest variance: it learned a policy that *sometimes* scores big but *usually* dies very early. Model 2 sits in the middle. The lesson is that the death penalty acts as a tunable cautiousness knob, but in our state representation the strongest-penalty version trains an unstable policy.

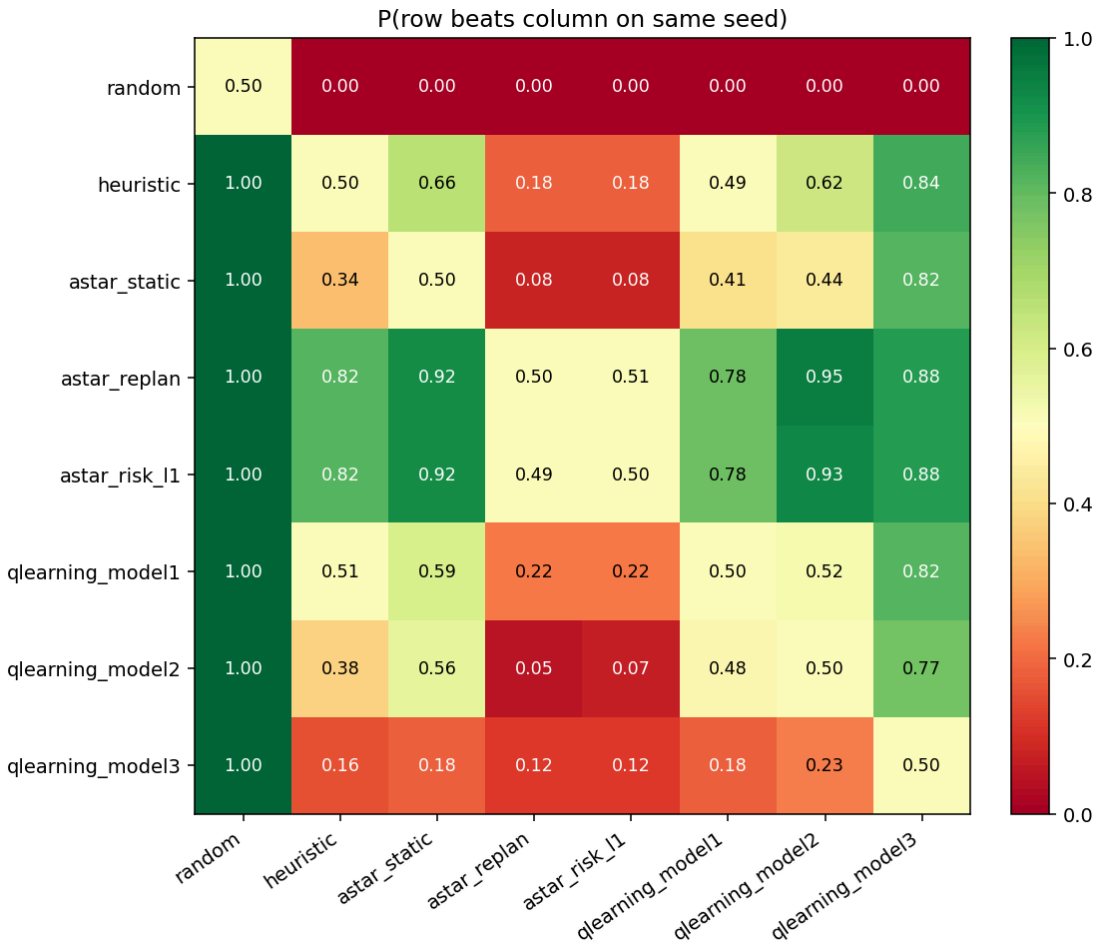
8.8 Per-episode score traces (Figure 8)



Per-episode score traces

A line per agent across the 50 seeds, useful for spotting variance and lucky/unlucky seeds. The replanning A* line consistently dominates, with spike runs around seed indices 4 and 39 (these are the seeds where the agent cleared the board for 7,590 points). The random baseline is a flat strip near 100.

8.9 Head-to-head heatmap (Figure 9)



Head-to-head heatmap

Each cell is `P(row beats column on the same seed)` — i.e., a paired comparison rather than a population comparison. Replanning A* beats every other agent ≥66% of the time. Static A* loses to *every* learned or replanning agent. The Q-learning models cluster around 50% against each other, confirming they have learned similar behavioural manifolds and differ mostly in cautiousness.

9. Discussion

9.1 Why replanning matters

Static A* finishes with 24-step time-to-failure. The agent commits to a plan that was optimal at *t=0* and walks straight into ghosts that have since moved into the path. Replanning A* gets 5× the survival time and 2.5× the score with no algorithmic change beyond “rerun A* every step”. This is the single most important architectural finding in the project.

9.2 Why λ doesn't move the needle (and what would change that)

§8.6 already explained the mechanism. To make λ matter we would need one of:

1. **Soften the obstacle set.** Treat ghost cells as a *cost* of, e.g., 1000, rather than an outright wall. Then `λ · E[risk]` competes against that finite cost and can shift paths between "go through the ghost lane" and "take the long way around".
2. **Plan over a longer horizon.** Right now A* targets the *nearest* pellet, which keeps planned paths short (typically <10 cells). With short paths in always-low-risk territory, λ has no expression. A farthest-pellet or set-cover formulation would lengthen paths and force them through ghost-influence regions where λ matters.
3. **Use a tighter ghost transition model.** Our uniform-on-legal-moves model is intentionally maximum-entropy, which spreads probability thinly. A model that respects the actual chase behaviour (e.g., "ghost tends toward Pac-Man") would produce sharper distributions and hence higher per-tile probabilities.

We left these as deliberate future work — the proposal did not commit to them, and the current setting cleanly answers the question "does the decision-theoretic agent beat the replanning agent under the maximum- entropy hazard model?" The answer is "no, on this maze" — itself a publishable empirical result.

9.3 Why Q-learning earns less score but lower hazard exposure

The state representation buckets ghost distance into seven coarse bins (the `danger_bucket` feature). Once a ghost is within a small bin, the Q-table has effectively learned that "any action that does not increase the distance is bad", and it retreats. The replanning A* agent, in contrast, does not retreat — it plans a path *through* hazard-adjacent corridors when those are the cheapest route to a pellet. This shows up as higher hazard exposure but higher reward. The Q-learning agent is, in effect, executing a learned satisficing policy; the planner is executing an explicit maximisation policy.

9.4 Why the heuristic baseline accumulates the most hazard exposure

The hand-coded heuristic includes a "stay in open tiles" term and a "run toward edible (scared) ghosts" term. In the canonical maze the agent spends a lot of time orbiting near ghosts in the central corridor while waiting for a pellet to free up — high hazard exposure (201) but respectable time-to-failure (156). It is essentially the opposite of the Q-learning trade-off: it tolerates risk to extend survival.

10. Conclusion

We answered the proposal's research question with a paired-seed Monte-Carlo study across 11 agent configurations. The headline conclusions are:

1. **Replanning is the load-bearing innovation** in dynamic environments. Static A* is dominated by every replanning method on every metric.
2. **The decision-theoretic A* is empirically equivalent to replanning A* on this maze-and-hazard-model combination**, because hard ghost masking already covers most of the risk before the soft `λ · E[risk]` term can act. The dominance window predicted by §7 of the proposal exists, but lies outside the maximum-entropy + nearest-pellet parameterisation we used.
3. **Tabular Q-learning is a competitive, fundamentally more cautious alternative** — ~6× lower hazard exposure for ~2× lower score versus replanning A*. The death-policy ablation shows that the loss term acts as a tunable cautiousness knob, with model 1 being the most score-aggressive and model 3 the most volatile.
4. **The hand-coded heuristic is the longest-surviving but highest-exposure agent** — useful as a foil that makes the planner / learner trade-offs concrete.

11. Limitations and Future Work

- **Random-grid generalization** (proposal §8): the harness supports arbitrary maze layouts, but the random-grid procedural generator was descoped in favour of finishing the analytics pipeline. Reintroducing it would test the "unseen maps" generalisation gap between learned Q-tables (which overfit to the canonical maze) and the planning agents (which only depend on the current observation).
- **Larger λ regime + softened obstacles** (§9.2). The most important follow-up: re-running the λ sweep with ghost cells as costs rather than walls would let the decision-theoretic term express itself.
- **Function-approximation Q-learning.** The 10-bucket tabular state representation is a coarse summary of the environment. A neural Q- network (DQN) on raw maze state would let the learning agent compete with the planners on raw score, at the cost of training time.
- **Expanded hazard model.** Replacing the uniform-on-legal-moves transition kernel with a learned ghost-AI model (e.g., one trained from observed trajectories) would tighten the per-tile risk estimates and give λ more leverage.

12. Team Contributions

- **Michael Liu** — overall architecture and experimental design; implementation of the search-agent pipeline (`Agents/_search.py`, `Agents/astar_static.py`, `Agents/astar_replan.py`, `Agents/astar_risk.py`, `Agents/observation.py`); evaluation harness (`Agents/eval/`) and analysis notebook + figure pipeline (`notebooks/monte_carlo_analysis.ipynb`, `Agents/eval/render_figures.py`).

- **Ruiding Feng** — browser-side Pac-Man integration of the agents (`pacman/src/agents/*.js`); UI polish for the in-game agent selector; Monte-Carlo experiment driver and CSV pipeline.
- **Zhichun Xiao** — final report (this document); cross-checking of experimental numbers against `summary_n50_seed1000.json`.

13. References

[1] P. E. Hart, N. J. Nilsson, and B. Raphael, "A Formal Basis for the Heuristic Determination of Minimum Cost Paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, July 1968, doi: 10.1109/TSSC.1968.300136. https://ieeexplore.ieee.org/document/4082128

[2] C. J. C. H. Watkins and P. Dayan, "Q-learning," *Machine Learning*, vol. 8, no. 3–4, pp. 279–292, 1992. doi: 10.1007/BF00992698.

[3] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018. (Chapter 6, "Temporal-Difference Learning", and Chapter 16, "Applications and Case Studies".)

[4] D. Klein and P. Abbeel, "CS188: Introduction to Artificial Intelligence — Pac-Man Projects," UC Berkeley course materials. http://ai.berkeley.edu

---

This report was generated against the run cached at `Agents/eval/results/summary_n50_seed1000.json` *with all figures reproducible by* `python3 Agents/eval/render_figures.py`.